

Matteo Leonelli

Cybersecurity Researcher

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 [/Leonelli](https://github.com/Leonelli)

 Italian: Native  English: C1  German: B2  Spanish: B1

PhD candidate in system security specializing in coverage-guided fuzzing, vulnerability research, and binary analysis, with peer-reviewed publications at top venues (NDSS, EuroS&P) and industry research experience at AWS. Discovered multiple CVEs across hardware and web platforms.

Experience

Amazon Applied Scientist Internship - AWS

Research Internship

Paris, FR

Sep 2025 – Nov 2025

- Designed and implemented automated fuzz testing pipelines to evaluate and ensure data integrity in AWS S3
- Applied grey-box fuzzing techniques to cloud storage systems, identifying edge-case failure modes in large-scale distributed infrastructure

CISPA Helmholtz Center for Information Security - Saarland University

Ph.D. Candidate

Saarbrücken, DE

Mar 2024 – Feb 2028

- **EuroS&P'26:** THEMIS — Context-Aware Grey-box Fuzzing for WordPress Plugins; discovered multiple authenticated/unauthenticated vulnerabilities in widely-deployed plugins
- **NDSS'25:** [TWINFUZZ](#) — Differential testing framework for video hardware acceleration stacks, leading to [Intel CVE-2024-23919](#)
- **SBFT'25:** [Fuzzing Track Tool Competition](#) at ICSE 2025
- **CVEs discovered:** [Intel CVE-2024-23919](#), [CVE-2025-9219](#), [CVE-2025-5701](#), [CVE-2023-7306](#)

Master's Thesis

Mar 2023 – Jan 2024

- Research on Software and Hardware Fuzzing in collaboration with Intel

Research Assistant

Apr 2022 – Mar 2024

- Collaborated with PhD researchers on automated root cause analysis of fuzzer-discovered bugs, investigating techniques to localize vulnerabilities to specific commits and code versions across software histories
- Researched LLM-driven automatic fuzzing harness generation to reduce manual effort in setting up fuzzing targets across different frameworks

Fondazione Bruno Kessler

Bachelor Thesis

Trento, IT

May 2021 – Sep 2021

- CODASPY'21 paper: [Secure Pull Printing with QR Codes and National eID Cards: A Software-oriented Design and an Open-source Implementation](#)

Junior Researcher

Feb 2019 – Dec 2020

- Designed authentication protocols using the Italian CIE 3.0 smart card standard

Software Developer Internship

Nov 2018 – Jan 2019

- Mobile developer internship in Security&Trust Lab

Java Developer Internship

Jun 2017 – Jul 2017

- Java developer involved with data analysis and software development in Smart Community Lab

Matrix Tech

Full Stack Developer

Trento, IT

Dec 2017 – Dec 2020

- Developed a full-stack web service for access control and real-time occupancy monitoring, integrated with proprietary embedded hardware

Education

Saarland University

Master in Cybersecurity

Saarbrücken, DE

Oct 2021 – Feb 2024

Trento University

Bachelor in Computer Science

Trento, IT

Sep 2018 – Sep 2021

Technological Institute "G. Marconi"

Diploma in Computer Science and Telecommunications

Rovereto, IT

Sep 2013 – Jul 2018

Summer Schools

SG, USA, CH, DE, IT

Fuzzing and Software Security Summer School 2026 - Intel Workshop: Scalable Assurance 2024 – SBFT 2025 – EPFL Summer School in System Security 2024 – CISP Summer School in System Security 2023 – CISP Summer School in Trustworthy Artificial Intelligence 2022 – University of Trento School of Innovation 2021 - Predictive Maintenance Summer School 2018

Teaching

Teaching Assistant

Nov 2022 – Apr 2024

- Prepared and delivered lectures, conducted exercise sessions, and created, administered, and graded exams for classes of 100+ students
- Saarland University (Nov 2022 – Apr 2024): Teaching Assistant in *Security, System Security, Human Computer Interaction* classes
- University of Trento (Sep 2020 – Feb 2021): Teaching Assistant in *C/C++ Programming* class

Bachelor Thesis Supervisor – Saarland University

Nov 2024 – Jul 2025

- Supervised a bachelor thesis on automated web application fuzzing targeting WordPress plugin vulnerability discovery
- Continued mentoring the student as Research Assistant (Oct 2025 – Present), overseeing day-to-day research activities and project direction

Technical Skills

Languages: C, C++, Python, Bash, Rust, PHP

Fuzzing & Security Tools: AFL++, libFuzzer, honggfuzz, WPScan, Burp Suite, GDB, Valgrind, AddressSanitizer

Research Areas: Coverage-guided Fuzzing, Differential Testing, Binary Analysis, Web Application Security, Hardware Security Testing

Platforms & Infrastructure: Linux, AWS, Docker, Git, QEMU